

TROUBLESHOOTING NETWORK SERVICES

1. **Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.**

1. Open Notepad as Administrator.
2. Navigate to `C:\Windows\System32\drivers\etc\hosts`.
3. Add entry:
Code: `123.123.123.123 www.spotify.com`
4. Save and close the file.
5. Open Command Prompt.
6. Type: `ping www.spotify.com`.
7. Observe the output → it resolves to wrong IP and shows “Request timed out” errors.

```
Windows IP Configuration
```

```
Successfully flushed the DNS Resolver Cache.
```

```
Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=59ms
Reply from 2a04:4e42:31::810: time=79ms
Reply from 2a04:4e42:31::810: time=38ms
Reply from 2a04:4e42:31::810: time=49ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 79ms, Average = 56ms
```

2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Configuration Steps

Step 1: Configure DHCP on Router

Open Router CLI:

```
enable
configure terminal
```

Code

Create DHCP pool with wrong gateway: Code

```
ip dhcp pool STUDENT
network 192.168.10.0 255.255.255.0
default-router 192.168.20.1 ← (Wrong Gateway)
dns-server 8.8.8.8
```

Assign IP to router interface:

```
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

Code

Step 2: Connect Client PC

- Go to **PC** → **Desktop** → **IP Configuration**
- Select **DHCP**
- The PC will receive:
 - IP: 192.168.10.x
 - Subnet Mask: 255.255.255.0
 - Default Gateway:

192.168.20.1 (wrong) For Verification

On PC Command Prompt:

Code

```
ipconfig /all
```

You will see the wrong gateway (192.168.20.1). Try pinging router:

Code

```
ping 192.168.10.1
```

Ping works, but internet/external communication fails because gateway is incorrect.

3. **Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.**

Troubleshooting Steps

1. **Check network connectivity** ping 8.8.8.8
 - If it works, the internet connection is fine.
2. **Test DNS resolution using nslookup**
nslookup www.google.com
OR (Linux)
dig www.google.com
3. **Check the result**
 - If it returns an IP address → DNS is working.
 - If it shows "**Server failed**", "**NXDOMAIN**", or "**DNS request timed out**", DNS is not working.
4. **Check the configured DNS server**
 - Windows: ipconfig /all
 - Linux: cat /etc/resolv.conf
5. **Change the DNS server**
Use a public DNS server such as:
 - Google DNS: **8.8.8.8** and **8.8.4.4**
 - Cloudflare DNS: **1.1.1.1** and **1.0.0.1**
6. **Clear the DNS cache**
 - Windows: ipconfig /flushdns
 - Linux: sudo systemctl restart systemd-resolved

(or restart the DNS service used by your system)
7. **Test again**
nslookup www.google.com
or
dig www.google.com

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

□

Possible Causes

1. Network connectivity problem – cable unplugged, wrong VLAN, or switch port down.
2. DHCP scope exhaustion – no free IPs left in DHCP pool.
3. DHCP server unavailable – router/server interface shutdown or service disabled.
4. Incorrect DHCP relay (IP helper) configuration – if DHCP server is on another subnet.
5. Firewall/ACL blocking DHCP traffic (UDP ports 67/68).

Troubleshooting Steps

Step 1: Check Physical Connectivity

- Verify cable between PC → Switch → Router.
- On PC:

Code

```
ipconfig /renew
```

If still APIPA → move to next step.

Step 2: Verify DHCP Service on Router/Server

On Router CLI:

Code

```
show running-config
```

- Confirm DHCP pool exists.
- Ensure interface connected to client has IP and no shutdown.

Step 3: Check DHCP Scope Availability

On Router:

Code

```
show ip dhcp binding
```

- If all IPs are leased → expand pool.
- Example fix:

Code

```
ip dhcp pool STUDENT
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

Step 4: Verify Client Settings

On PC → Desktop → IP Configuration:

- Ensure DHCP is selected, not static.
- Release/renew:

Code

```
ipconfig
/release
ipconfig
/renew
```

Step 5: Check DHCP Relay (if server on another subnet)

On Router interface:

Code

```
ip helper-address <DHCP-server-IP>
```

Without this, DHCP requests won't reach server.

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Type	issue	Symptoms	Possible Causes	Recommended Solutions
DNS	Wrong DNS Server	Can ping IP but not domain	DHCP pool misconfigured	Correct DNS server in DHCP config, renew client IP
DNS	DNS Server Down	nslookup timeout	Server off / firewall block	Start DNS service, check firewall
DNS	Cache Corruption	Some sites fail intermittently	Local DNS cache issue	Flush DNS cache
DHCP	Scope Exhaustion	New devices get APIPA	Pool full	Increase DHCP scope range
DHCP	Server Down	Clients fail to get IP	Router interface shutdown	Restart interface, DHCP service
DHCP	Relay Misconfig	Remote subnet clients fail	Missing/wrong helper address	Configure correct

